



INTEGRATED SCIENCE STAGE 3 DRAFT SAMPLE EXAMINATION

Section 7 of the New WACE Manual: General Information (Revised) outlines the policy on WACE examinations.

Further information about the WACE Examinations policy can be accessed from the Curriculum Council website at <http://www.curriculum.wa.edu.au/>.

The purpose for providing a sample examination is to provide teachers with an example of how the course will be examined. Further finetuning will be made to this sample in 2008 by the examination panel following consultation with teachers, measurement specialists and advice from the Assessment, Review and Moderation (ARM) panel.

DRAFT



**Western Australian Certificate of Education, Draft Sample External
Examination
Question/Answer Booklet**

**INTEGRATED SCIENCE
WRITTEN PAPER
STAGE 3**

Please place your student identification label in this box

Student Number: In figures

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In words

Time allowed for this paper

Reading time before commencing work: 10 minutes

Working time for paper: 3 hours

Materials required/recommended for this paper

To be provided by the supervisor

Question/answer booklet

Integrated Science: Formulae and Data Sheet (inside front cover of this Question/Answer Booklet)

Separate multiple choice answer sheet

To be provided by the candidate

Standard items: Pens, pencils, eraser or correction fluid, ruler, highlighter

Special items: Non-programmable scientific calculator, drawing instruments and templates.

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Suggested working time (minutes)	Number of questions available	Number of questions to be attempted	Marks available
Section One: Multiple-choice	30	20	20	40 (20%)
Section Two: Short answer	90	6	6	100 (50%)
Section Three: Extended answer	60	2	2	60 (30%)
Total marks				200 (100%)

Instructions to candidates

1. The rules for the conduct of the Western Australian external examinations are detailed in the *WACE Handbook*. Sitting this examination implies that you agree to abide by these rules.

SECTION ONE: MULTIPLE-CHOICE*[40 marks]*

There are **TWENTY (20)** questions in this section. Attempt **ALL** questions.

Mark your answers to questions 1–20 on the SEPARATE MULTIPLE CHOICE ANSWER SHEET using a 2B, B or HB pencil.

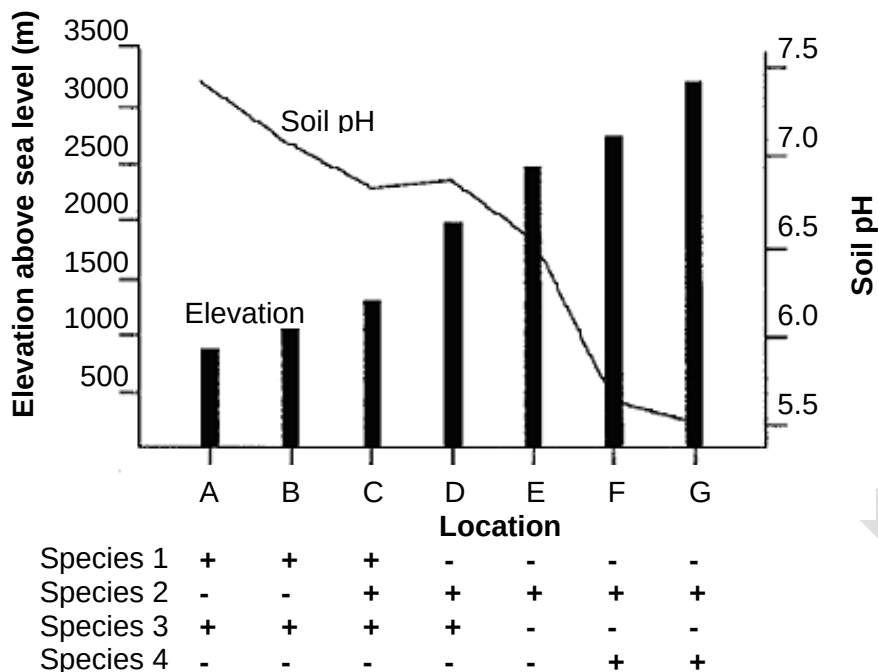
Suggested working time: 30 minutes

1. The process used in the extraction of mineral sands could best be described as
 - (A) open cut mining.
 - (B) underground mining.
 - (C) strip mining.
 - (D) bench mining.
2. Before mining can commence in an area a company is required to submit an Environmental Impact Statement, the purpose of which is to
 - (A) assess the potential adverse and beneficial environmental, economic and social impacts of the project.
 - (B) identify only the benefits to society and employment that are likely to come from the proposal.
 - (C) construct a cost benefit analysis statement of the proposal and see if it is financially viable.
 - (D) only identify the environmental costs of the project.
3. After a mining project has been completed the company is required to rehabilitate the area. Rehabilitation involves
 - (A) planning quick growing trees to prevent erosion.
 - (B) planting trees appropriate to the area.
 - (C) re-establishing the ecosystem.
 - (D) spreading seeds of plants collected locally.
4. Bioleaching is the
 - (A) use of fine membranes to remove micro-organisms from an ore.
 - (B) use of water percolation to remove micro-organism from an ore.
 - (C) use of micro-organisms to extract metals from an ore.
 - (D) use of micro-organisms to make the metals ready for extraction.

Use the following data for questions 5, 6 and 7

The diagram below shows data from an ecological transect. It indicates which species are present at each location from A to G and shows the elevation and soil pH at each location.

Effect of elevation and soil pH on distribution of species



Key:

Present = +

Absent = -

5. Which location contains the least and the greatest diversity of species?

Least

Greatest

- (A) B
(B) E
(C) C
(D) A

- F
C
E
G

6. Which species is found over the least and the greatest range in the soil pH?

Least

Greatest

- (A) Species 3
(B) Species 2
(C) Species 4
(D) Species 1

- Species 1
Species 4
Species 2
Species 3

[From: Australian Science Olympiads, 2003]

7. The relationship between elevation and pH could be described as

- (A) soil pH is constant as the elevation decreases.
(B) soil pH decreases as the elevation decreases.
(C) soil pH increases as the elevation increases.
(D) soil pH decreases as the elevation increases.

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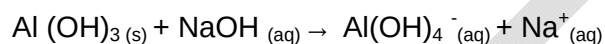
8. Tin metal is electroplated onto sheets of iron using a tin anode and a well-stirred solution containing Sn^{2+} ions. During this process

- (A) the anode increases in mass.
- (B) Sn^{2+} ions move towards the cathode.
- (C) the concentration of Sn^{2+} in the solution decreases.
- (D) the concentration of Sn^{2+} in the solution increases.

9. The use of carbon in the production of iron involves the

- (A) loss of electrons by oxygen.
- (B) loss of electrons by iron.
- (C) gain of electrons by carbon.
- (D) gain of electrons by iron.

10. In the Bayer Process, a reaction occurs between the aluminium hydroxide in the solid ore and the hydroxide ion to produce the aluminate complex ion.



What is the role of the sodium hydroxide?

- (A) To dehydrate the aluminium hydroxide.
- (B) To separate the soluble materials from the insoluble materials.
- (C) To produce aluminium in solution.
- (D) To produce aluminium hydroxide.

11. Nuclear fusion is the process where

- (A) heavy nuclei are split resulting in the release of energy.
- (B) heavy nuclei are split resulting in the absorption of energy.
- (C) light nuclei are joined together resulting in the release of energy.
- (D) light nuclei are joined together resulting in the absorption of energy.

12. Heat engines are devices which

- (A) convert heat into useful work.
- (B) convert work into useful heat.
- (C) convert heat into electricity.
- (D) convert electricity into heat.

13. Which of the following is not a fossil fuel?

- (A) natural gas
- (B) coal
- (C) uranium
- (D) oil

14. Current, potential difference and resistance are represented by the symbols

- (A) I, V and R respectively.
- (B) R, V and I respectively.
- (C) C, V and R respectively.
- (D) P, V and I respectively.

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15. The outer bricks of a house get hot on a sunny day because of

- (A) convection.
- (B) evaporation.
- (C) conduction.
- (D) radiation.

16. How many units of electricity will a 1830W small electric fan heater use if it is left on for 3 hours?

- (A) 5.5 J
- (B) 5.5 W
- (C) 5.5 kWh
- (D) 5.5 V

17. How can energy and work both have the same unit, i.e. 'Joule'?

- (A) Work has a lower case j as the symbol and energy has a capital J.
- (B) Energy has a lower case j as the symbol and work has a capital J.
- (C) Energy is the change in work, so energy is the work before and work after a process.
- (D) Work is the change in energy, so work is the energy before and energy after a process.

18. Biogas is the general name given to methane which is produced by the biological decomposition of materials such as

- (A) animal wastes including sewage.
- (B) rubbish from landfill sites.
- (C) left over fibre from sugar cane processing.
- (D) all of the above.

19. Power can be defined as the

- (A) ability to do work.
- (B) rate of change of energy.
- (C) rate of change of work.
- (D) ability to use energy.

20. Renewable sources of energy are sustainable sources of energy if

- (A) their original source of energy is the sun.
- (B) they are renewed quicker than they are consumed.
- (C) they can be used to produce electricity.
- (D) they are based on plants.

END OF SECTION ONE

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SECTION TWO: SHORT ANSWER AND PROBLEM SOLVING*[100 marks]*

There are **SIX (6)** questions in this section. Attempt **ALL** questions.

Suggested working time: 90 minutes

Question 1

An iron ore mine using an open pit operation is proposed for the Pilbara region near Newman in Western Australia has been approved on the basis of an environmental impact assessment

A summary of the project is provided below.

Element	Description
Life of project	4 years
Ore reserves	12 million tonnes
Ore mining rate	2–4 million tonnes per annum
Pit depth below existing plain level	140 metres
Pre-mining water table level below plain level	3 metres
Pit area	32 hectares
Overburden storage area	105 hectares
Total area disturbed	137 hectares
Water abstraction (removal)	38,000 kilolitres per day (max)
Area within the 10m drawdown	360 hectares

[From: Department of Resources Development, 1999]

(a) Explain the significance of the pit depth and the water table level figures.

[2 marks]

(b) Explain why changes in the amount of water removed per day may occur as the depth of the pit increases over the life of the mine.

[2 marks]

- (c) Water pumped out of the mine is used for dust suppression around the mine site and on mining access roads, with any excess water pumped into a nearby creek.

Describe the effect that operating this mine would have on the surface and groundwater within the 360 hectare area associated with the mine.

[2 marks]

Surface water: _____

Groundwater: _____

- (d) The Pilbara region has a semi-arid climate. The local river and its creek systems support trees such as the River Red Gum which depends on a seasonal source of water. Other vegetation in the area depends on either groundwater or seasonal rainfall. Aquatic organisms live in the permanent and semi permanent pools. Fauna of the area include birds, reptiles, amphibia, large and small marsupials, and a variety of aquatic and terrestrial invertebrates. Subterranean fauna include several categories of fauna associated with a range of underground habitats. These habitats vary in void size and type, from dry to fresh or brackish water.

Explain how changes in the surface and groundwater during the life of the mining operation would affect the following ecology of the area.

- (i) Flora

[2 marks]

- (ii) Fauna

[3 marks]

- (iii) Subterranean fauna

[1 mark]

(e) An environmental impact assessment of a mining project must also take into account the short and long term effects of the mining operations after the mine has closed. One of the concerns is that, in the long term, saline lakes will be formed in the disused pits.

(i) Explain how water levels are predicted to change once the mining operation has ceased and the pit has been abandoned.

[1 mark]

(ii) How will the inflow rate of water and the rate of evaporation affect the salinity of the water in the disused mining pit?

[2 marks]

(f) Mining companies use many petroleum-based products in the operation of equipment used in mining. For example, hydraulic fluids are used in earth-moving machinery. Why is it important to account for all waste products produced from mining operations?

[1 mark]

Question 2

It is increasingly difficult to locate new ore deposits near the Earth's surface that are suitable for mining. They currently search for ore deposits below the Earth's surface, which are sometimes at great depths.

Successful mineral exploration requires a well-planned program to pinpoint likely areas of deep mineral deposits. An exploration program involves the work of a team of geologists, geophysicists and geochemists.

- (a)** Geologists use ground-mapping techniques to identify features seen on satellite images and aerial maps of large tracts of continents. Explain why mineral exploration starts with this process.

[3 marks]

- (b)** When a promising site has been identified, geophysicists measure the gravity, magnetic and electrical properties of the rocks.

Explain how **one** of these methods (gravity, magnetic, electrical properties) is used to indicate the presence of a mineral deposit.

[5 marks]

- (c) Geochemical surveys typically follow a series of steps from examination of large areas to progressively smaller and smaller areas until analysis and geological assessment has located a deposit. Types of geochemical surveys may include stream sediment surveys, rock sampling and soil sampling.

Describe each of the following methods.

- (i) Stream sediment surveys

[2 marks]

- (ii) Rock sampling

[1 mark]

- (iii) Soil sampling

[2 marks]

- (a) A further geochemical survey sometimes carried out is trenching, which is a traditional method of exposing concealed bedrock. Trenching is normally undertaken using an excavator. Describe the rehabilitation involved once the process is completed.

[2 marks]

- (d) Under the *Environmental Protection Act 1994*, an Environmental Impact Statement (EIS) needs to be prepared for a mining project if the Environmental Protection Agency or the Minister for the Environment decides it is appropriate. One purpose of the EIS is to assess management, monitoring, planning and other measures proposed to minimise any adverse environmental impacts of the project.

Describe measures that would be included in the EIS to protect the vegetation within the 360 hectare area affected by mining operations.

[3 marks]

Question 3

The mining of bauxite and production of aluminium involves processes that impact on health and the environment.

- (a) Describe how the environment is altered in preparation for the mining of bauxite.

[2 marks]

- (b) It has been suggested by some that mining of bauxite improves the jarrah forest while others have suggested that 'only the trees are replaced and not the forest'.

What is meant by the statement 'only the trees are replaced and not the forest'?

[3 marks]

- (c) Soil quality is the key factor in stable and sustainable restoration of land disturbed by mining. The quality of soil can be measured in terms of physical, chemical and biological fertility.

Explain the change to the soil profile that occurs as a result of bauxite mining.

[2 marks]

- (d) *Phytophthora cinnamomi* is a fungal disease affecting many native trees found in the jarrah forest. It grows into the root systems preventing the absorption of water and nutrients. In warm and moist conditions, spores are released into the surrounding soil, where they can be spread to other trees through soil water movement.

Explain how changes in the soil profile due to bauxite mining may limit the spread of *Phytophthora cinnamomi*.

[2 marks]

- (e) When the alumina is removed from the bauxite ore, the remaining impurities are discarded. This waste product is known from its oxidised iron content as 'red mud', and because it has been subjected to sodium hydroxide treatment, is highly corrosive with pH values in excess of 13.2. Consequently it gives rise to storage problems and is an on-going environmental liability.

State the main problem associated with wet disposal methods for the red mud.

[1 mark]

- (f) Residue neutralisation will help reduce the potential for environmental impacts from residue storage activities. The cost of neutralisation will be offset by a reduction in the need for long term management of the residue deposits.

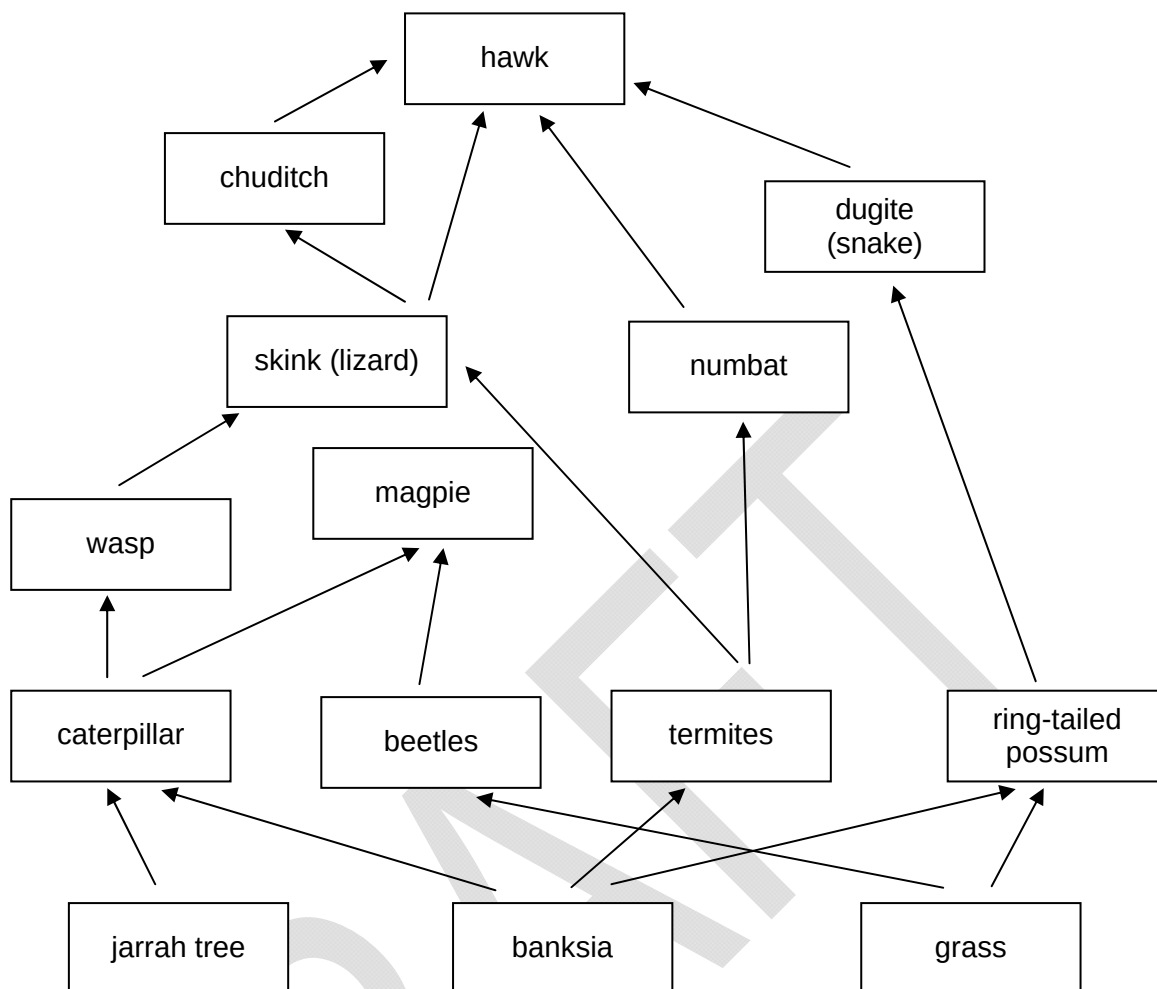
One method of neutralisation that has been developed and tested in Western Australia is carbonisation (carbon dioxide treatment).

Describe the benefits of residue carbonisation.

[3 marks]

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(g) The diagram below shows a food web for jarrah forest in the Darling Range.



Name all of the foods of the magpie in this food web.

[2 marks]

(h) The food web consists of several different food chains.

(i) From the food web, draw a food chain containing six organisms.

[1 mark]

(ii) Explain why it is rare for food chains to involve more than five organisms.

[2 marks]

Question 4

Biomass is a form of renewable energy which is derived from a variety of animal and plant materials, including waste products and crops, as a result of photosynthesis.

(a) Define photosynthesis.

[1 mark]

(b) Explain how biomass energy derived from animals can be due to photosynthesis.

[1 mark]

(c) To be environmentally sustainable, the chemical energy obtained from a biomass source must be greater than the energy expended in growing and processing a biomass source. Explain why.

[3 marks]

- (d) One process for converting biomass into useful energy is to change animal and plant waste into biogas. Explain how biochemical processes can be used to produce methane from organic wastes. Name two sources of biogas in your explanation.

[4 marks]

- (e) Ethanol has been used as a motor fuel at various times in the past. Describe one advantage and one disadvantage of using ethanol as a fuel, or a fuel supplement.

[2 marks]

Advantage: _____

Disadvantage: _____

- (f) Explain how combustion of biomass sources can be used to produce electricity.

[3 marks]

Question 5

In the future, it seems likely that hydrogen, a zero-carbon fuel, will be used more predominantly as an energy source. Some of this hydrogen will be generated by electrolysis of water using renewable energy sources such as hydro, wind or solar power.

(a) Describe, using a diagram, the production of hydrogen from water by electrolysis.

[4 marks]

(b) The production of hydrogen using electrolysis is less than 100% efficient, therefore hydrogen is not considered a primary source of energy.

As using electricity to produce hydrogen uses energy, why would hydrogen be considered for use as a fuel for vehicles?

[2 marks]

- (c) Electricity can be generated by burning hydrogen in a power plant or by other methods.

Describe an alternative method of generating electricity from hydrogen.

[2 marks]

- (d) Suggest four difficulties that might be encountered in using hydrogen rather than petroleum as a fuel for vehicles.

[4 marks]

Question 6

Incandescent light globes will be phased out in Australia from 2009, as a way of reducing Australia's energy use and greenhouse gas emissions, by encouraging the use of compact fluorescent lights (CFL).

- (a) You replaced a 100W incandescent globe with a 100W equivalent CFL, with an 18W rating. If your bedroom light was on four (4) hours per day, how much energy would you save in 1 year?

[4 marks]

- (b) How much energy will be saved each year if 6 incandescent 100W globes are replaced by 18W CFLs in 8 million households in Australia?

[2 marks]

- (c) List four practical ways to conserve energy in an existing home other than by replacing incandescent globes with CFLs.

[4 marks]

- (d) Use your knowledge of heat transfer to describe how each of the following contributes to an energy efficient home.

- (i) Orientation of the house and location of windows

[2 marks]

(ii) Building materials

[3 marks]

(iii) Design of the building

[3 marks]

(e) Energy is often wasted in our homes. Energy is lost through windows, walls, roofs, up chimneys, through ventilation ducts and around doors. List four practical ways through which these losses may be reduced, other than those mentioned in part d.

[4 marks]

END OF SECTION TWO

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Question 8

Figure 1 below shows the current available energy sources.

Figure 2 shows the predicted energy supply by each type if there are no changes to current policy.

Figure 3 shows an alternative scenario based on policy changes to be considered.

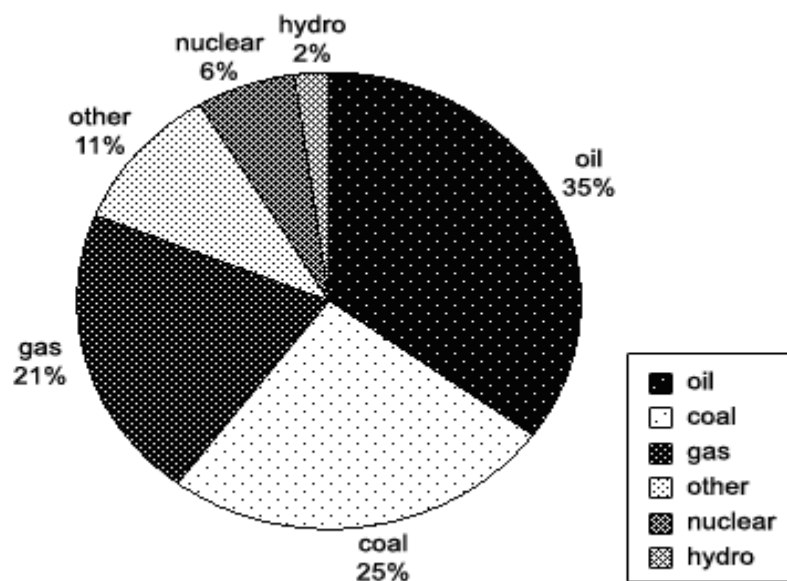
The predicted total energy supply is given in megatonnes of oil equivalent (Mtoe) for each scenario. The 'other' energy sources category shown includes biofuels, solar, geothermal, tidal and wind.

Discuss the global impacts of each of the predicted scenarios in figures 2 and 3. Include in your answer explanations of:

- how current energy sources compare with predicted energy sources in 2030
- how fossil fuels compare with alternative energy sources
- the impact of changes in carbon dioxide emission
- advantages and disadvantages of using **two** of the 'other' energy sources shown in the figures
- advantages and disadvantages of using nuclear power.

[30 marks]

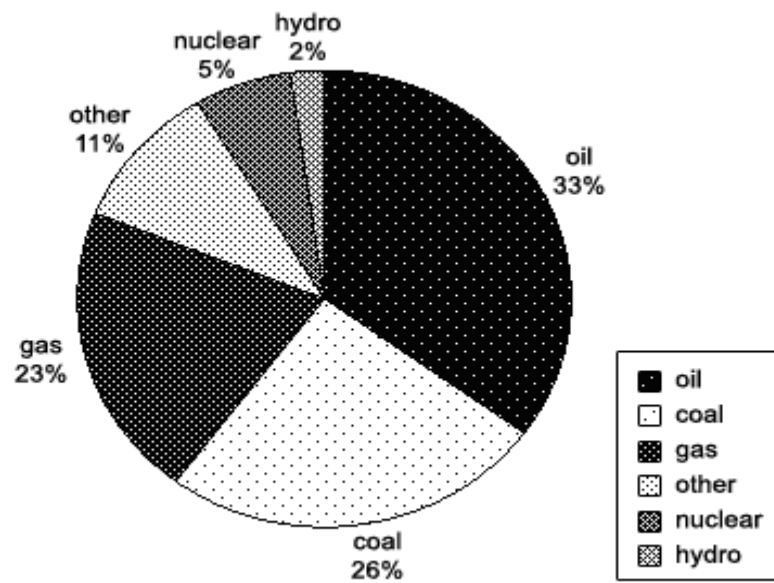
Figure 1: Current Energy Sources



Total 11 453 Mtoe

[From: International Energy Agency, 2005]

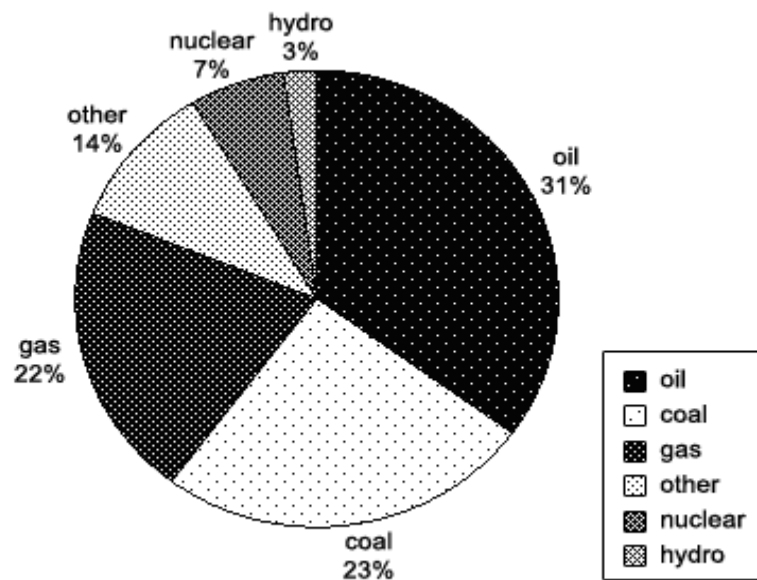
Figure 2: Predicted Energy Sources 2030
(current policies)



Total 17 100 Mtoe

[From: International Energy Agency, 2007]

Figure 3: Predicted Energy Sources 2030
(alternate scenario based on policy changes)



Total 15 400 Mtoe

[From: International Energy Agency, 2007]

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ACKNOWLEDGEMENTS

SECTION ONE—MULTIPLE CHOICE:

Question 5 & 6: Adapted from: Australian Science Innovations. (2003). *Biology: 2003 National Qualifying Examination* (p. 11). Retrieved November, 2007, from Australian Science Olympiads website:
<http://www.aso.edu.au/www/docs/2003%20%20Biology%20NQE%20Questions.pdf>.

SECTION TWO—SHORT ANSWER AND PROBLEM SOLVING:

Question 1: Department of Resources Development. (1999). *Mining below the water table in the Pilbara* (p. 21). Retrieved November, 2007, from
http://www.ecc.online.wa.gov.au/documents/environment/shed_env_about_miningbelowwatertablepilbara.pdf.

SECTION THREE—EXTENDED ANSWER:

Question 8: **Figures 1**
Adapted from International Energy Agency. (2005). *Key world energy statistics* (p. 46). Retrieved November, 2007, from
<http://www.iea.org/textbase/nppdf/free/2005/key2005.pdf>

Figures 2 & 3
Adapted from International Energy Agency. (2007). *Key world energy statistics* (p. 46). Retrieved November, 2007, from
http://www.iea.org/Textbase/nppdf/free/2007/Key_Stats_2007.pdf

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